Brain-Inspired Fast Saliency-Based Filtering Algorithm for Ship Detection in High-Resolution SAR Images

Panpan Zhang, Haibo Luo, Moran Ju, Miao He, Zheng Chang, and Bin Hui

***Abstract*— In this article, we aim to improve the performance of synthetic aperture radar (SAR) ship detection under complex conditions. The complex backgrounds are commonly encountered for high-resolution (HR) SAR ship detection data set, and they greatly influence the detection performance of ships. In recent years, deep neural networks (DNNs) have made substantial improvements on detection by adopting data augmentation. However, the improvement is limited since the models are sensitive to noise. To address this problem, a Fast Saliency-based Filtering algorithm (FSF) is proposed to filter out interference information. The FSF method is inspired by the filtering mechanisms of the human brain, which help people filter out target-irrelevant information fast to better extract target-relevant information. The FSF includes two parts of the bottom-up process and the top-down process. The bottom-up process is used to extract a saliency map of an input image, and the other one is used to filter out target-irrelevant information based on the saliency map. The FSF can be a front-end preprocessing module of DNNs to fast filter out target-irrelevant information and obtain a primary priority map of an input image. Experimental results demonstrate that our brain-inspired FSF method obtains obvious improvement of detection performance on AIR-SARShip-1.0.**

***Index Terms*— Brain-inspired, deep neural networks (DNNs), Fast Saliency-based Filtering algorithm (FSF), filtering mechanisms, priority map, saliency map.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging radar that can generate high-resolution (HR) images in all-weather conditions; thus, the SAR images have been widely used in a variety of civil and military applications. Ship detection is an important application in the field of marine surveillance [1]. Due to the complexity and uncertainty of marine weather conditions, the SAR images are more suitable for ship detection. SAR ship detection has gained popularity in recent years [2].

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Due to the ability of SAR system, many solutions are proposed to process SAR images, such as wavelet transform [3], sparse representation [4], the Gabor filter [5], bag-of-words (BoW) [6], and POLWHE (PolSAR image segmentation based on LMDL, ZWNFB, and HMRF-EM) [7]. With the development of deep learning in recent years, deep neural networks (DNNs) can extract high-level features, and many methods based on DNNs are proposed to be used in SAR images [8]–[11]. In fact, ship detection in satellite applications is often processed on the whole scene image, covering an area of tens of square kilometers or more. Under such conditions, the environment around the target, such as docks, roads, auxiliary buildings, and even waves, also has a great impact on the ship detection performance [12].

Deep learning has brought on remarkable improvements in object detection. Object detectors are usually categorized into two kinds: one-stage object detectors and two-stage object detectors. One-stage detectors mainly include You Only Look Once (YOLO) [13]–[16], single-shot multibox detector (SSD) [17], and RetinaNet [18]. Two-stage object detectors are mainly the region convolutional neural network (R-CNN) [19] series, including Fast R-CNN [20], Faster R-CNN [21], region-based fully convolutional networks (R-FCNs) [22], and Libra R-CNN [23]. In recent years, anchor-free one-stage object detectors are developed, such as CornerNet [24], fully convolutional one-stage object detection (FCOS) [25], and CenterNet [26].

Although object detectors based on CNNs have realized state-of-the-art results on widely used benchmark data sets, their performance tends to be weaker on near-shore ship detection due to the features extracted by convolutional modules that are affected easily by noises.

To address this problem, many solutions have been proposed. The most commonly used method is data augmentation, such as random noise, flipping, rotating, and contrast enhancement. In addition to these ways, more complex and deeper networks are used to extract target features and obtain better detection performance. However, these operations improve detection performance with the cost of computation consumption. To address this problem, brain-based learning is becoming popular, and now, the advance of neural science and cognitive neuroscience can also provide a more comprehensive understanding of the human brain. So far, there are two main bits of intelligence: machine intelligence (MI) and artificial

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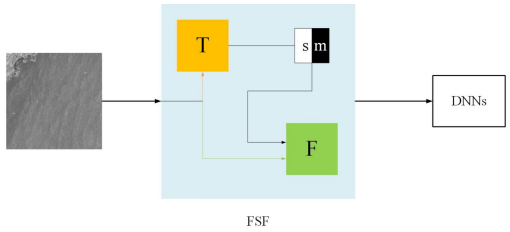


Fig. 1. Framework of DNNs with FSF.

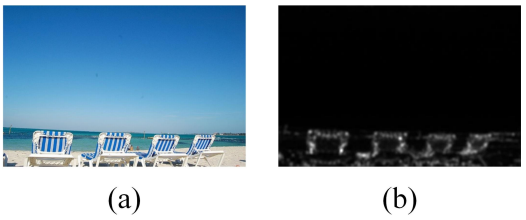


Fig. 2. (a) Original image. (b) Saliency map.

intelligence (AI). In electronic engineering, the MI is used to develop a machine vision system, while AI is used in computer engineering. AI has been a hot topic since it was proposed. In history, the great breakthrough of deep learning mainly gives the credit to the brain's mechanisms. For example, LeCun *et al.* [27] proposed a convolution neural network based on the visual receptive field of cat, Rivest *et al.* [28] proposed Brain-like reinforcement learning based on basal ganglion, Serre *et al.* [29] proposed Hierarchical Model and X (HMAX) module based on the inferior temporal cortex of the monkey brain, and Sabour *et al.* [30] proposed a capsule network based on cortical microcolumn structure. Besides, the widely used attention module [31] is also proposed based on the attention mechanism of the brain and has obtained remarkable performance in both the computer vision field and the natural language processing (NLP) field.

In this work, we propose a Fast Saliency-based Filtering algorithm (FSF) based on the filtering mechanisms of the human brain. The FSF is used to filter out complex background information rapidly, such as lands, buildings, and trees, before images are fed into DNNs, which can reduce the influence of noises on SAR ship detection, as shown in Fig. 1. The HR SAR ship detection data set (AIR-SARShip-1.0) [12] is used in our experiment. Experimental results demonstrate our superiority.

The remainder of this article is organized as follows. Section II gives an introduction of methodology. In Section III, application to HR SAR ship detection is introduced in detail. Experimental results and analysis on AIR-SARShip-1.0 are described in Section IV. Finally, Section V gives the conclusion.

II. Methodology

The filtering mechanisms include two kinds of types: focusing directed by a goal and focusing directed not by a goal. Focusing directed by a goal is the more conscious top-down process, while focusing directed not by a goal is the automatic bottom-up process. The two processes generate the brain's internal representation of its environment—priority map [32]–[34].

A. Bottom-Up Process

The automatic bottom-up process filters out the information below the level of awareness. In this case, our focus is directed not by a goal but rather by particular properties of the stimuli, such as their brightness or motion. There are different conclusions about the place where the automatic bottom-up process takes place, such as the parietal cortex [35]–[37], frontal eye field [38], and superior colliculus [39]. The more important spatial distribution map produced by the automatic bottom-up process is called the saliency map, as shown in Fig. 2.

There are a lot of relative studies about extracting saliency map, such as IT (ITTI) [40], graph-based visual saliency (GBVS) [41], frequency-tuned (FT) salient region detection [42], and region-based contrast (RC) [43]. The most representative module is IT, integrating color and brightness, and direction information in different scales. With the development of deep learning in recent years, saliency detection algorithms based on DNNs are also proposed, such as DeeFix [44], SALICON [45], Mr-CNN [46], eDN [47], and DVA [48].

In this work, we propose a threshold method to obtain the saliency map of the HR SAR ship image, according to its brightness information. This part is detailed in Section III.

B. Top-Down Process

The top-down process is a kind of attentional way by which the brain suppresses irrelevant stimuli. For instance, we disregard greens, blues, and yellows in a crowd when we are searching for our friends wearing red. In this case, our focus is directed by a goal. Wimmer *et al.* [49] introduced a new subcortical model of sensory selection, in which the prefrontal cortex (PFC) biases thalamic reticular subnetworks to control thalamic sensory gain, selecting appropriate inputs for further processing. Nakajima *et al.* [50] proposed that the PFC instructs the basal ganglia (BG) to employ the thalamic reticular nucleus (TRN) that dynamically controls thalamic gain through feedforward inhibition. Furthermore, the filtering mechanism does not just filter out one sense but also filters information within a single sense, as illustrated in Fig. 3.

To express the recognition principle intuitively, we use abstract mathematical linguistics to illustrate the filtering algorithm of different sensory information and the filtering algorithm within a single sense, as defined by (1) and (2), respectively

$$\mathcal{I}_{V\downarrow,A} = \mathcal{I}_{V,A} \ominus \mathcal{I}_V \tag{1}$$

where $\mathcal{I}_{V,A}$ represents the information of visual stimulus and auditory stimulus, $\mathcal{I}_V$ represents the target-irrelevant information of visual stimulus, $\ominus$ represents the inhibiting effect, and $\mathcal{I}_{V\downarrow,A}$ represents the remaining information

$$\mathcal{I}_{A_\alpha\downarrow,A_\beta} = \mathcal{I}_{A_\alpha,A_\beta} \ominus \mathcal{I}_{A_\alpha} \tag{2}$$

where $\mathcal{I}_{A_\alpha,A_\beta}$ represents the information of auditory stimuli, $\mathcal{I}_{A_\alpha}$ represents target-irrelevant information of auditory

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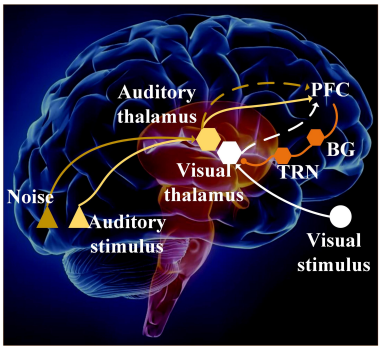


Fig. 3. Top-down process mechanism of the human brain. Here, we assume that the auditory stimulus is the target, so the target-irrelevant information, such as visual stimulus and the auditory noise, is filtered out before they reach the PFC.

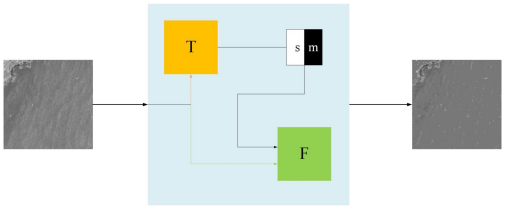


Fig. 4. FSF. T represents the threshold method. sm represents a saliency map extracted by T. F represents the filtering algorithm.

stimulus, $\ominus$ represents the inhibiting effect, and the $\mathcal{I}_{A_\alpha\downarrow,A_\beta}$ represents the remaining information.

In this work, we propose a filtering algorithm inspired by this filtering mechanism of the human brain. This part is detailed in Section III.

III. Application to High-Resolution SAR Ship Detection

In this work, according to the bottom-up process and the top-down process, we propose an FSF. The FSF includes two parts: the threshold method and the filtering algorithm, as shown in Fig. 4.

A. Threshold Method

According to the image-forming principle of SAR images, the brightness information is its main feature. Thus, we use the threshold method to realize the bottom-up process and extract its saliency map. All images are normalized to [0, 1] first, and then, the threshold T $\in$ (0, 1) is selected to extract the saliency map by (3). The threshold method not only can achieve real-time performance but also extract salient features of SAR images, as shown in Fig. 5. The effectiveness is consistent with the bottom-up process of the human brain

$$f(x) = \begin{cases} 0 & x < T \\ 255 & T \le x. \end{cases} \tag{3}$$

The threshold T is fairly important. It determines which features are salient and which are not, and it also has an impact on the result of the filtering algorithm. If T is too large, we would lose excessive information about the complex background; if T is too small, the target information would be influenced by the complex background, as shown in Fig. 6. In order to reserve the entirety of the target object, T is set to 0.7 experimentally.

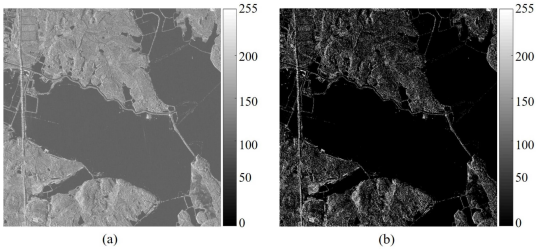


Fig. 5. (a) SAR ship image. (b) Saliency map.

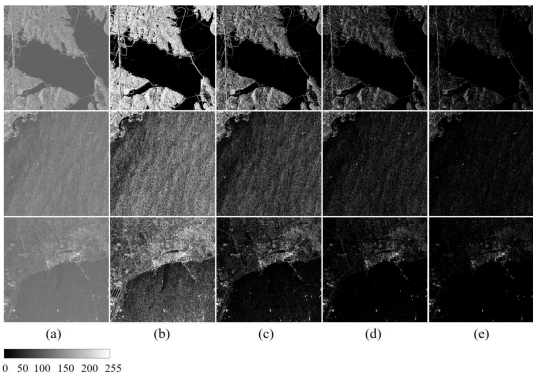


Fig. 6. SAR ship images and saliency maps under a different threshold T. (a)–(e) Original images, T = 0.5, T = 0.6, T = 0.7, and T = 0.8, respectively. All subimages are in gray scale and with the same range. The range of color bar is from 0 to 255.

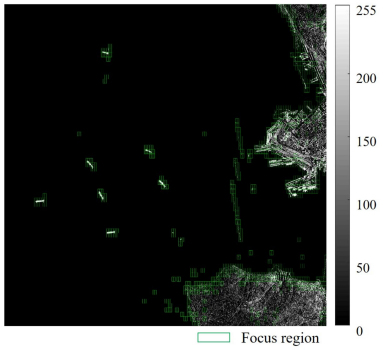


Fig. 7. Green box represents the focus region.

B. Filtering Algorithm

According to *a priori* knowledge that ships usually appear in far-sea and near-shore areas, the far-sea and near-shore areas should be the focus areas in SAR ship images, as shown in Fig. 7. In other words, the irrelevant and complex screen should be filtered, such as lands, buildings, and trees, as defined in the following equation:

$$\mathcal{I}_{V_{\text{trees},\ldots,\text{lands}}\downarrow,V_{\text{coast,sea}}} = \mathcal{I}_{V_{\text{image}}} \ominus \mathcal{I}_{V_{\text{trees},\ldots,\text{lands}}}. \tag{4}$$

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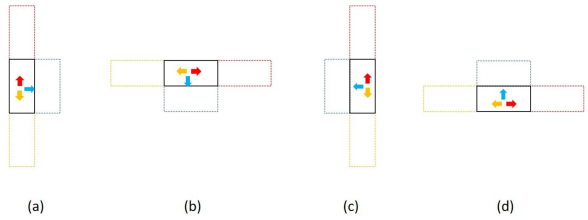


Fig. 8. Multidirectional contrast strategy based on anchor. (a)–(d) Four kinds of the movement pattern of anchor: start with left of the image, start with the top of the image, start with the bottom of the image, and start with the right of the image, respectively. The black box represents the anchor region. Dotted boxes represent the forward region (dotted blue box), the upward region or right region (dotted red box), and the downward region or left region (dotted yellow box), respectively. Arrows represent different movement directions of the anchor.

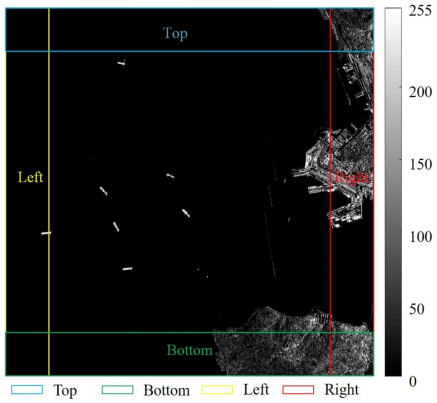


Fig. 9. Decision of movement direction of anchor. The top, bottom, left, and right regions are used to compute respective pixel density.

In order to filter out complex background information as much as possible and meanwhile reserve the entirety of ships, we propose the FSF method that utilizes a multidirectional contrast strategy based on the anchor, as shown in Fig. 8(a). First, we determine the movement pattern of anchor via pixel density, as shown in Fig. 9. In other words, the anchor starts with one side of image that contains complex background. Second, we count the number of pixels of current anchor region, then move the anchor forward, up, and down, and count the number of pixels of these regions, respectively. Finally, we compute the ratio of the number of pixels of anchor region and the number of pixels of other regions in different directions by (5), respectively. The ratio r determines whether the anchor region should be filtered

$$r = \frac{\text{Min}(\sum \mathcal{A}_{\text{region}}, \sum \mathcal{O}_{\text{region}})}{\text{Max}(\sum \mathcal{A}_{\text{region}}, \sum \mathcal{O}_{\text{region}})} \tag{5}$$

where r represents the similarity of pixel density between anchor box and other boxes in different directions, $\sum \mathcal{A}_{\text{region}}$ represents the number of pixels of anchor region, and $\sum \mathcal{O}_{\text{region}}$ represents the number of pixels of one of other regions in different directions.

Algorithm 1 describes how the proposed method works for an HR SAR ship image. For each input image, we first get its saliency map according to the threshold T in Line 1. As described in Lines 4, 6, 12, and 21, $\mathcal{L}$ and $\mathcal{D}$ are used to keep the anchor box inside the image. The current anchor region $\mathcal{A}_{\text{region}}$ is initialized in Line 10 and determines whether its forward region should be reserved in Lines 11–20. After

Algorithm 1 FSF

Input:
$\mathcal{I}_{original}$ is an original image
$\mathcal{W}$ is the width of image
$\mathcal{H}$ is the height of image
T is the threshold of saliency map
w is the width of filtering box
h is the height of filtering box
$\mathcal{R}$ is the initial ratio

Output:
$\mathcal{I}_{filtered}$ is a filterd image

1: get saliency map according to T;
2: initialize horizontal stride distance $\mathcal{L} \leftarrow 0$;
3: initialize vertical stride distance $\mathcal{D} \leftarrow$ h;
4: **while** $\mathcal{D} \leq \mathcal{H}$ **do**
5: $\mathcal{L} = \mathcal{L}$ + w;
6: **if** $\mathcal{L} > \mathcal{W}$ **then**
7: $\mathcal{D} = \mathcal{D}$ + h;
8: start from next row $\mathcal{L} \leftarrow$ w;
9: **end if**
10: current anchor region in image $\mathcal{A}_{region} \leftarrow \mathcal{I}_{original}[\mathcal{D}\text{-h}{:}\mathcal{D},\mathcal{L}\text{-w}{:}\mathcal{L}]$;
11: **if** pixel number of $\mathcal{A}_{region}$: $\sum \mathcal{A}_{region} = 0$ **then**
12: **if** $\mathcal{L}$ + w $\leq \mathcal{W}$ **then**
13: **if** next region pixel number of $\mathcal{A}_{region}$: $\sum \mathcal{I}_{original}[\mathcal{D}\text{-h}{:}\mathcal{D},\mathcal{L}{:}\mathcal{L}\text{+w}] > 0$ **then**
14: save current anchor region $\mathcal{A}_{region}$;
15: continue;
16: **else**
17: change background;
18: **end if**
19: **end if**
20: **end if**
21: **if** $\mathcal{L}$ + w $\leq \mathcal{W}$ **then**
22: get next region of $\mathcal{A}_{region}$: $\mathcal{N}_{region} \leftarrow \mathcal{I}_{original}[\mathcal{D}\text{-h}{:}\mathcal{D},\mathcal{L}{:}\mathcal{L}\text{+w}]$;
23: **if** $\sum \mathcal{N}_{region} = 0$ **then**
24: save current anchor region $\mathcal{A}_{region}$;
25: continue;
26: **else**
27: compute ratio of $\sum \mathcal{N}_{region}$ and $\sum \mathcal{A}_{region}$: r = Min($\sum \mathcal{N}_{region}$,$\sum \mathcal{A}_{region}$) / Max($\sum \mathcal{N}_{region}$, $\sum \mathcal{A}_{region}$)
28: **if** r $< 1 / \mathcal{R}$ **then**
29: save current anchor region $\mathcal{A}_{region}$;
30: continue;
31: **else**
32: change background;
33: **end if**
34: **end if**
35: **end if**
36: The operation of other directions of upward and downward is as same as the forward direction.
37: **end while**
38: return $\mathcal{I}_{filtered}$

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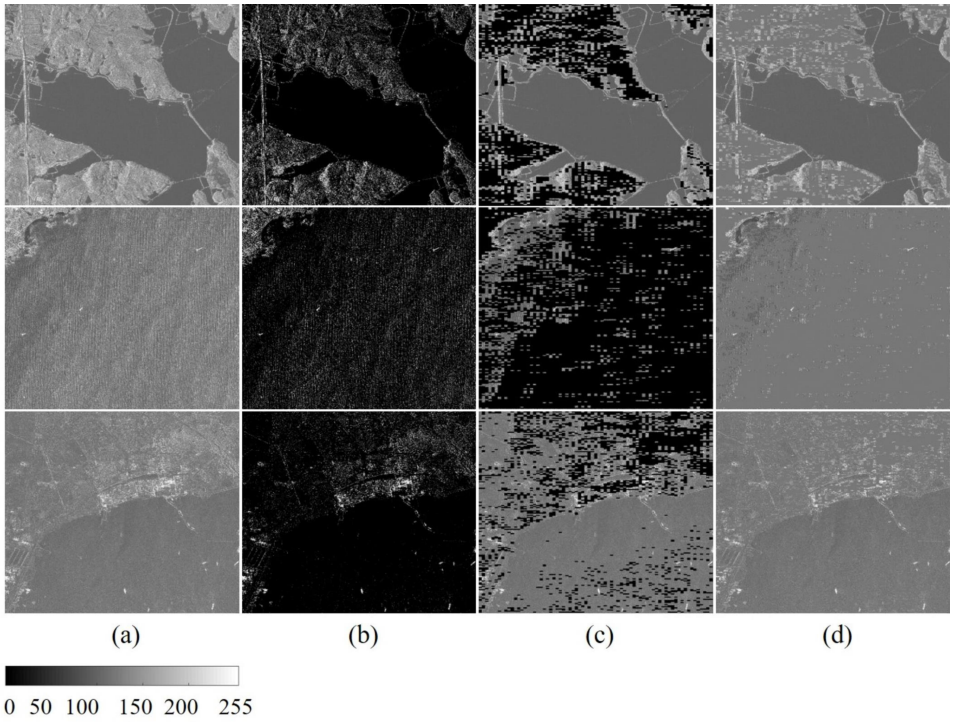


Fig. 10. Three different scenes are shown. (a) Original image. (b) Saliency map. (c) Primary priority map. (d) Filtered image. All subimages are in grayscale and with the same range. The range of color bar is from 0 to 255.

that, we determine whether the current anchor region should be reserved, according to the forward direction region in Lines 21–35 and the upward direction region and the downward direction region in Line 36, respectively. It is worth noting that the parameter $\mathcal{R}$ is a constant and is set to 2.5 in our experiments. In addition, the size of the anchor box is a critical parameter. If it is too large, an amount of information of complex background will be reserved; inversely, ships will be destroyed. We determine the anchor size based on the size of the ship in HR SAR ship images.

Different scenes and their processed results are shown in Fig. 10. The primary priority map reserves target-relevant information, as shown in Fig. 10(c). To reserve scene information of sea, we replace complex background information with a single sea background information, as shown in Fig. 10(d).

IV. Experiments and Analysis

A. Data Sets for Experiments

We conducted experiments to evaluate the detection performance of the proposed FSF, and the AIR-SARship-1.0 data set is used. The AIR-SARShip-1.0 data set based on Chinese GF-3 satellite [51] includes 31 different scene images. Its resolutions are 1 and 3 m, and the imaging mode includes the spotlight and stripmap modes. The polarization mode is single polarization, the image format is Tiff, and the size of most images is 3000×3000, as listed in Table I. The AIR-SARship-1.0 data set prominently includes two types of the scene: near-shore areas and far-sea areas, as shown in Fig. 10(a). Single background information is a prominent characteristic of the far-sea areas. However, the near-shore areas include wharves, buildings, and lands, which results in lower detection performance of the ships. The detailed information of the AIR-SARShip-1.0 data set is listed in APPENDIX.

TABLE I
Data Set Information

Resolution	Imaging mode	Polarization mode	Image format
1m, 3m	spotlight, stripmap	single	Tiff

B. FSF Detection Results and Analysis

1) Experimental Setup: In the AIR-SARship-1.0 data set experiments, the original images are processed by the FSF first and then cropped into 500×500. In addition, some augmentation methods, such as randomly deforming the brightness (increasing pixel value, and the value is 32), contrast (the lower bound of the contrast factor is 0.5, and the upper is 1.5), rotation (the angle of rotation is 90 °), and horizontal flip operation (the probability of flip is 0.5), are used to augment data. We select classical one-stage object detector, SSD, and two-stage object detector, Faster-RCNN for comparison. In the SSD experiments, the input image size is scaled into 512×512. The hyperparameters are as follows: the batch size is 12; the base learning rate is 0.000004; and the others are the same as default settings. In the Faster-RCNN experiments, we use an ImageNet pretrained weight VGG16.v2.caffemodel and then utilize the training model of end-to-end. We use default hyperparameters, and the input size is 500×500. All experiments use Convolutional Architecture for Fast Feature

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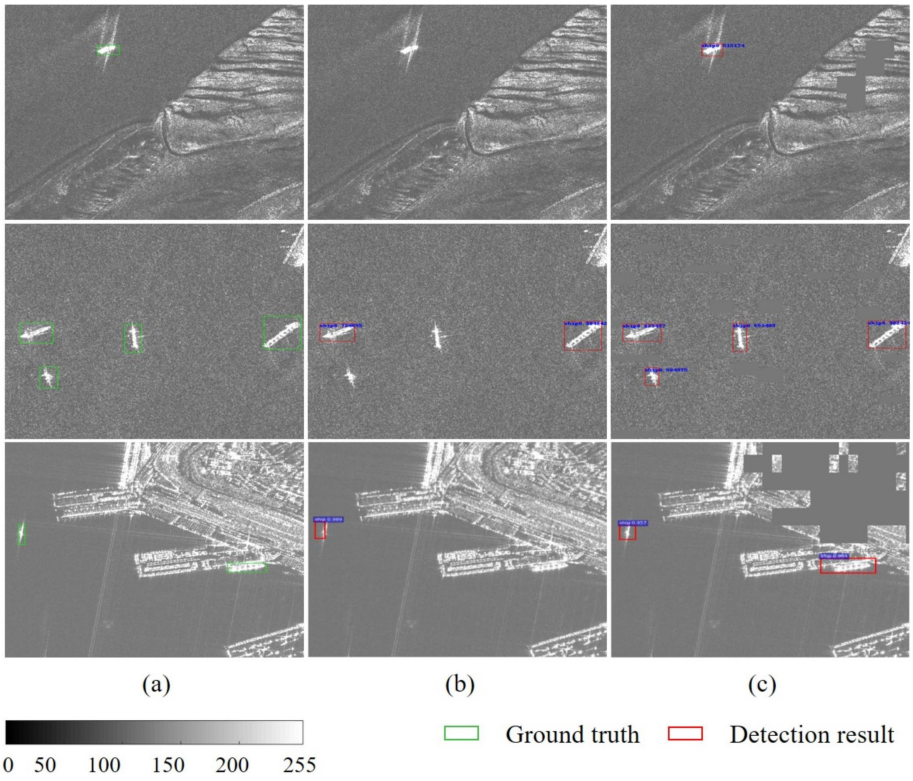


Fig. 11. Detection results. (a) Green boxes represent the ground truth. (b) Red boxes represent the detection results of DNNs. (c) Red boxes represent the detection results of our method. All subimages are in gray scale and with the same range. The range of the color bar is from 0 to 255.

Embedding (CAFFE) and are trained with a TITAN×12-GB graphics processing unit (GPU).

2) Detection Results and Analysis: The time complexity of FSF is $\mathcal{O}(n)$, and the speed of running is relative to the size of the image. In our experiment, the FSF consumes $\sim$ 300ms to process an image of 3000×3000 sizes. Comparison results of detection are shown in Table II. Because the AIR-SARship-1.0 data set contains only one class of ships, the mean Average Precision (mAP) for all classes is the same as the average precision (AP) value.

The AP scores for different scenes intuitively indicate the performance of baseline models and the FSF model, as shown in Table II. The FSF improves the performance of near-shore areas, outperforming the SSD model and Faster-RCNN model by 11.9% points and 5.2% points, respectively. The results demonstrate the effectiveness of the proposed method, and some visual detection results are shown in Fig. 11. Some far-sea ships cannot be detected by the baseline model, as shown in Fig. 11(b), whereas the FSF can effectively detect the ships of far-sea areas and near-shore areas, as shown in Fig. 11(c). It is worth noting that the accuracy of FSF is 0.2% points lower than that of Faster RCNN on far-sea ship detection. As for that result, we think that some ships are damaged.

TABLE II
COMPARISONS WITH BASELINE OF DNN MODEL

Method	Input size	AP(%)	$AP_{nearshore}$(%)	$AP_{far-sea}$(%)
SSD	512×512	80.2	44.9	86.6
FSF+SSD(**Ours**)	512×512	**80.9**	**56.8**	**87.5**
Faster-RCNN	500×500	67.6	25.1	77.1
FSF+Faster-RCNN(**Ours**)	500×500	**68.1**	**30.3**	76.9

V. CONCLUSION

In this work, we propose an FSF module inspired by the filtering mechanisms of the human brain. The proposed module can filter out target-irrelevant information fast and reserve target-relevant information in the meantime. On AIR-SARship-1.0, the FSF fully utilizes its features to realize the separation of the target region and interference region. The FSF can also combine neural networks almost without a cost of consumption of time. Experiment results on the AIR-SARship-1.0 data set demonstrate the superior performance of our method with respect to baseline results of neural networks.

This article mainly concentrates on reserving the near-shore region, the far-sea region, and the integrality of foreground targets. In the future, further experiments will be conducted to filter out target-irrelevant information as much as possible, such as some wharves and rocks. We believe that a better filtering algorithm can further improve detection accuracy.

APPENDIX

See Table III.

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TABLE III
DETAILED DATA SET INFORMATION

Image number	Image size	Sea conditions	Scene	Resolution(m)	Number of ship
1	3000×3000	Level 2	nearshore	3	5
2	3000×3000	Level 0	nearshore	1	7
3	3000×3000	Level 3	far-sea	3	10
4	3000×3000	Level 2	far-sea	3	8
5	3000×3000	Level 1	nearshore	3	15
6	3000×3000	Level 4	far-sea	3	3
7	3000×3000	Level 4	far-sea	3	5
8	3000×3000	Level 1	nearshore	1	2
9	3000×3000	Level 2	nearshore	1	7
10	3000×3000	Level 1	far-sea	1	50
11	3000×3000	Level 1	nearshore	1	80
12	3000×3000	Level 2	nearshore	1	18
13	4140×4140	Level 1	nearshore	1	21
14	3000×3000	Level 1	nearshore	1	15
15	3000×3000	Level 1	nearshore	1	77
16	3000×3000	Level 3	nearshore	3	13
17	3000×3000	Level 3	nearshore	3	3
18	3000×3000	Level 3	nearshore	3	2
19	3000×3000	Level 3	nearshore	3	1
20	3000×3000	Level 2	nearshore	3	7
21	3000×3000	Level 2	nearshore	3	9
22	3000×3000	Level 1	nearshore	3	14
23	3000×3000	Level 1	far-sea	3	4
24	3000×3000	Level 4	far-sea	3	6
25	3000×3000	Level 4	far-sea	1	20
26	3000×3000	Level 2	nearshore	3	15
27	3000×3000	Level 2	nearshore	3	19
28	3000×3000	Level 1	nearshore	3	8
29	3000×3000	Level 3	far-sea	3	6
30	3000×3000	Level 2	far-sea	3	8
31	3000×3000	Level 1	nearshore	3	3

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